

The formation of a KARST LANDSCAPE

Karst landscapes are formed by the weathering of a terrain's underlying carbonate rock, such as limestone or dolomite. Through a process known as dissolution, rain, surface water, and groundwater made acidic by carbon dioxide in the atmosphere and organic matter in soil start to dissolve the rock down cracks and faults, and along its bedding planes. As the rock degrades over time and the cracks become larger, a variety of surface features are produced and an underground drainage system may start to form.

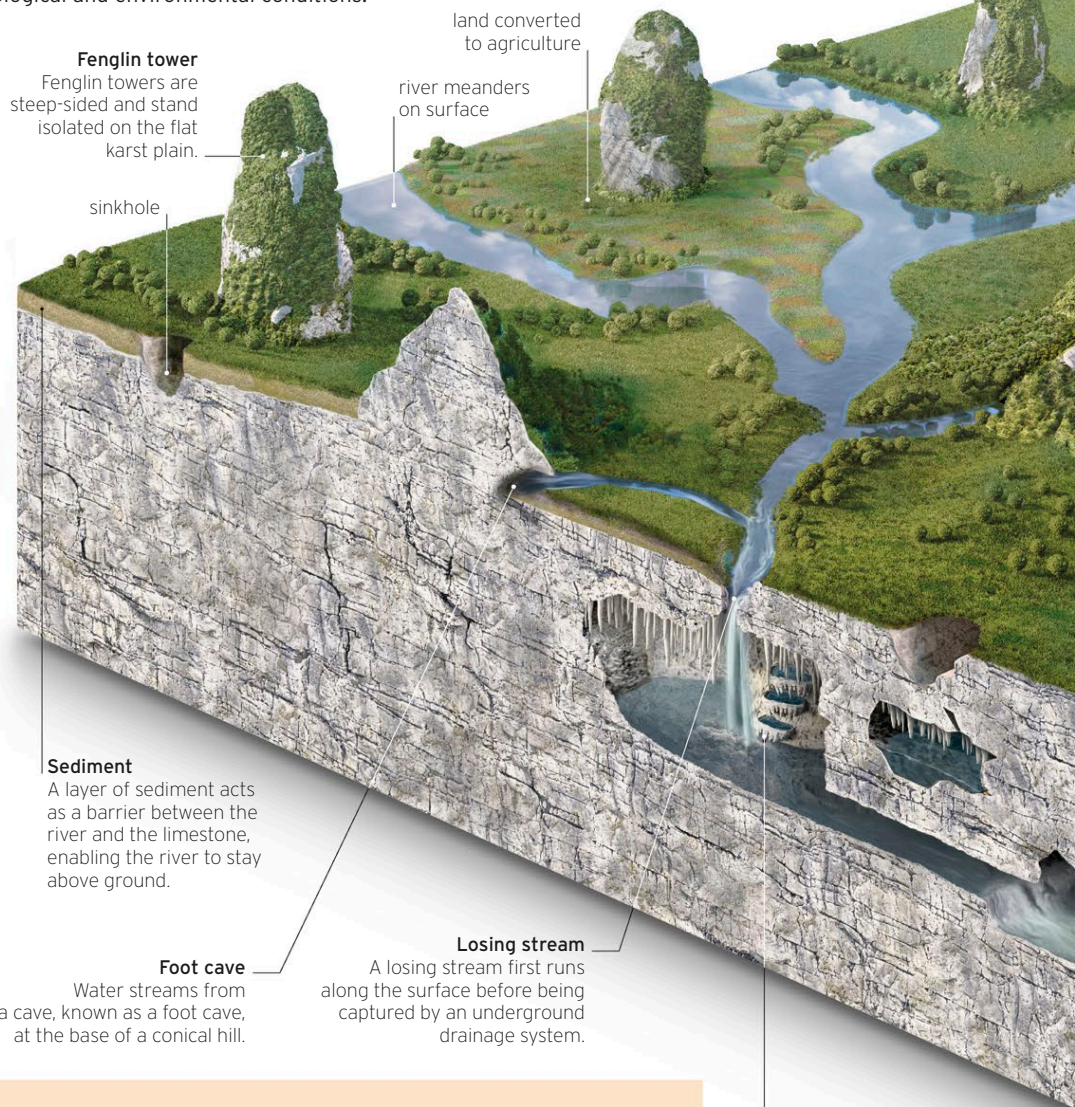
Underground erosion

Percolating water may create subterranean rivers that eventually carve out extensive cave systems. Sinkholes, or dolines, a defining feature of karst landscapes, either develop gradually as surface openings enlarge or are formed suddenly by cave collapses. As they expand and merge, they create sunken regions, and any high ground left between them becomes linked cone-shaped hills, such as the fengcong formations of South China karst. Over time, erosion by surface water can transform these hills into the isolated towers of fenglin karst. The formation of karst landscapes is helped by moist climates and lush vegetation—providing carbon dioxide as it decays—so the most developed karst terrains occur in tropical regions.

The **bedrock** of South China karst **formed on the seabed** more than **250 million years ago**

► SOUTH CHINA KARST

This cross-section of a karst landscape combines many of the features present in South China karst. It represents a well-developed karst system that has been created under perfect geological and environmental conditions.

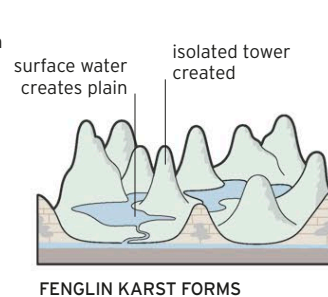
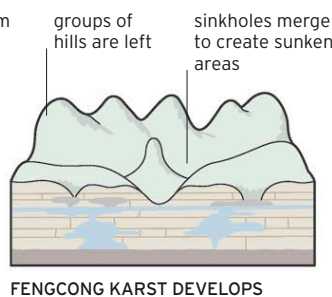
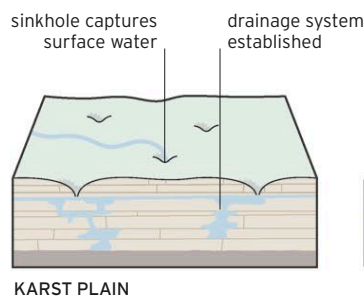


△ CAVE DWELLER

Caves provide shelter for some animals. Large colonies of great roundleaf bats roost in South China karst caves.

EVOLUTION OF KARST FEATURES

Karst features can be seen to evolve in a sequence from a flat karst plain to fengcong then fenglin karst. Once a drainage system is established, sinkholes get bigger and join together, producing the hills of fengcong karst. If surface water erodes their bases and the sunken areas between, they develop into the isolated towers of fenglin karst.



Gour
A series of stone dams called gours, formed by mineral deposits, has created a tiered cascade pool.